Investment Analysis: IB valuation methods

Written by: Nicolás Rivera Mendoza – Analyst | December

The purpose of investment banking

Investment banks guide their clients to decide how to raise or manage capital for their companies. Hence, Investment Banking (IB) is noted for its active role in mergers & acquisitions (M&A), share issuances, bond issuances, and initial public offerings (IPOs) of companies. However, the true value of IB to its clients comes from its access and creation of information to achieve the best deals in the previously mentioned operations, which is only possible with companies' valuation.

Valuation indicates how much a company is worth to others in the present and justifies acquisition or sale of proprietorship. This biased nature leads to different views on the same firm, and this depends on investors' objectives and methods of assessment. In IB valuation is used to determine the price of a company for a potential buyer or the market. In this way, IB does not only sets the monetary value of a company but analyzes how its development could generate value for investors.

How is valuation used in investment banking?

IB analysts consider the composition of capital structure (how much of debt and equity is being used to finance the business), the prospect of future earnings, and assets' market value to estimate valuation for companies. This information comes together in methods of valuation known as intrinsic and relative.

Valuation approaches in investment banking:

Intrinsic Valuation

With intrinsic valuation, a company's value is based on its fundamentals such as its cashflows, its growth and its risk. Discounted Cashflow (DCF) is the most common method used to find intrinsic value. DCF is based in companies cash flows from model projections.

DCF analysis assumes that the markets make mistakes in valuing individual companies and that they correct these mistakes over time. Additionally, it takes an indefinite amount of time for the market to reach the DCF valuation.

Therefore, the accuracy of valuation using DCF increases as the time horizon for the market to reach this valuation lengthens.

DCF states that the value of a company is the expected cashflows of that asset. In order to arrive to DCF value, a common five-step process can be followed.

1. The first step is gathering information. This process is fairly easy if the target company is public. If private, IB firms would typically be give this information if they are in the sell-side, while they might have to do due diligence for it if they are on the buy side.
2. The second step is projecting Free cash Flow (FCF). "Free cash Flow is the cash generated by a company after paying all cash operating expenses and taxes (...) but prior to the payment of any interest expense" (Pearl & Rosenbaum, 2022). FCF is calculated over a typical timeframe of five to ten years since its accuracy decreases as the timeframe lengthens.
3. The third step is calculating the Weighted Average Cost of Capital (WACC). WACC is a common way to determine the required rate of return for all the capital providers of the targeted company, weighted on the provider's portion in the capital structure (combination of debt and equity to finance company's operations). Therefore, WACC serves to comprehend the financial risk of companies and is often used as discount rate to find Present Value (PV). The discount rate is risk adjusted, meaning that safer assets have smaller discount rates than riskier assets.
4. The fourth step is determining Terminal Value. A terminal value is used to calculate the remaining value of the target company after the projection period. The Terminal value shows a steady financial performance into perpetuity unlike cyclical highs and lows.
5. The fifth step is determining Present Value (PV) and Enterprise Value (EV). EV is found by the summation of each projected FCF year's discounted Present Value (the value of the future cashflows in the present), plus the terminal value's discounted PV. Since DCF is built on assumptions, it offers a spectrum of the value of the company rather than an accurate number. To figure out the size of the spectrum, sensibility analysis is executed by changing key inputs such as WACC or Terminal Value.

For example, investment banking would find the intrinsic value of Toronto Dominion Bank (TD) in the following way:

Given that TD is a publicly traded company, IB analysts could only look at its cashflow statements. For the purpose of this example, I will consider the last four yearly consolidated cashflows statements in CAD which at the date of this paper are October 10th, 2024, 2023, 2022 and 2021.

The FCFs are displayed in Image 1. In IB, future FCF are projected by comparing inflows and outflows of cash. In this example, I will consider the change of total expenses (outflows) and the change in total revenue (inflows) to determine the ratio of increment of future cashflows. This assumption generates a -6% growth of future cashflows (average inflows = 8%; average outflows =14%) which are shown in image 2.

Investment Analysis: IB valuation methods

Written by: Nicolás Rivera Mendoza – Analyst | December

Image 1: *FCF, Total Expenses, ∆ Total Expenses, Total revenue, ∆Total revenue.*

*Total expenses were 29,379,000 and total revenue was 41,065,000 for 10/31/2020 TD income statement

	30/10/2024	30/10/2023	30/10/2022	30/10/2021
FCF	52,760,000,000	-67,146,000,000	37,495,000,000	49,000,000,000
Total expenses	45,115,000,000	38,597,000,000	25,338,000,000	25,132,000,000
Change in Net Expenses	17%	52%	1%	-14%
Total Revenue	55,945,000,000	51,485,000,000	45,762,000,000	42,266,000,000
Change in Total Revenue	9%	13%	8%	3%

Image 2: *Future FCF*

	30/10/2028	30/10/2027	30/10/2026	30/10/2025
Future FCF	41,537,469,696	44,096,713,685	46,813,640,118	49,697,964,271

To calculate the WACC, it would be considered the market capitalization and debt of TD. This information is publicly available in its balance sheets and is displayed in Image 1. Then, the cost of equity (Re) (i.e., the minimum rate required for the investment to be worth the risk) would give the value of 8.6% and the cost of debt (Rd) (i.e., the rate of interest the firm pays for all of its debt) would be of 14.9733%.

Image 2:

Market Cap (E): 91.443 B

Debt + Equity (V): 369.07 B

Debt (D): 277.627 B

TD Tax rate (Tc): 25% (ratio between

$$\text{WACC} = \left(\frac{E}{V} \times Re\right) + \left(\frac{D}{V} \times Rd \times (1 - Tc)\right)$$

$$\text{WACC} = \; 10.5955\%$$

In order to calculate the Terminal Value (TV), IB uses a perpetual growth rate (g) that should be close to the firm's country inflation rate, but lower than the country's gross domestic product (GDP) growth rate. The latter is because otherwise the perpetual rate would mean that the firm will do better than the entire country's economy for perpetuity.

Considering that Canada's average inflation rate and GDP growth since 1962 are 2.03% (Bank of Canada, 2024) and 3.04 respectively, I will use 2.53% as the perpetual growth rate to calculate the TV pf TD as shown on image 3.

Image 3:

Year 4 FCF (FCF_n): 41,537 B **WACC (r):** 10.5955 %

$$TV = \frac{FCF_n}{(r-g)}$$

$$TV = 491{,}246{,}957{,}324.43$$

It is important to mention that the PV of terminal value could even be worth 75% of the present value of the firm, given that it's the valuation of the business beyond the explicit period. In this case, the projected TV represents 67% of the intrinsic value of TD.

To find the DCF of TD, the next four-year cashflows' PVs are summed to the TV's PV which is found using the Gordon's growth model. To find DCF-derived share price, DCF is divided by the 1750058939 shares outstanding that TD has issued, giving a valuation of 252 CAD per share.

Image 4:

4-year FCF's PV *= 143,572,826,472.00*

TV's PV *= TV/(1+r) ^4*

TV's PV *= 288,706,802,823.38*

DCF *= 296,901,409,157.46 + 143,572,826,472.00= 432,279,629,295.38*

IB analysts would then compare the evaluated price to the current stock price of 75.03 CAD, determining that TD is undervalued by 70% of its true intrinsic value.

Relative Valuation

Relative valuation considers similar businesses and uses their multiples as a framework to analyze the target company. Multiples are used since they represent a scaled measure of price. Given that the values of similar firms can vary for being larger or smaller, multiples represent the ratio between same-category data about them and makes them comparable.

Comparable company analysis (COMPS) and Precedent Transaction analysis (PTA) are the most know approaches within this method of valuation. In COMPS, the analyzed company is compared with companies that are similar in size, product, and geography. This method of valuation is based on the idea that companies with similar characteristics should have similar valuation multiples. COMPS utilizes multiples from the latest stock prices and financials as a measure of comparison. Higher multiples of peers indicates that the evaluated company is potentially undervalued. Conversely, if the peers' multiples are lower, it could mean that the company is overvalued.

On the other hand, PTA asses relative value by looking at multiples of historical transactions (Pignataro, 2013) called purchase multiples. Through this analysis, the value assigned to the assessed company is relative to what it has been paid for other similar ones. Such as in COMPS analysis, if the company's purchase multiples are lower than its peers, it may be considered undervalued; if higher, it may be overvalued.

In IB, the peer universe is often instructed by the Managing Director or the deal team of the firm. Also, the management team of the target company would contribute by pointing at the true competition in the marketplace, and therefore it is usually the best resource. However, the selection is always decision of the investment banker.

For this example, I will continue to utilize TD to illustrate the IB approach to arrive to the relative valuation using COMPS. IB would utilize the comparative approach that is most appropriate for the target company. therefore, I will not be using PTA to find the relative value of TD given that there is no record of an acquisition of a commercial bank with the size of TD in Canada. Taking this into account, the following companies in image 5 share enough similarities to make them appropriate subjects of comparison.

Image 5: *COMPS for TD (Financial information from Yahoo Finance)*

	Market Data			Financial Data			Valuation	
	Price	Market Cap	Enterprise value (TEV)	Revenue	EBITDA	Net Income	EV/EBITDA	P/E
Bank of Montreal (BMO)	139	1.01E+11	3.66E+11	3.20E+10	9.33E+09	6.93E+09	39	15
CIBC (CM)	93	8.72E+10	2.17E+11	2.55E+10	9.09E+09	6.85E+09	24	13
National Bank of Canada (NA)	132	4.49E+10	1.21E+11	1.14E+10	4.77E+09	3.66E+09	25	12
Royal Bank of Canada (RY)	173	1.71E+11	7.28E+11	5.75E+10	1.99E+10	1.59E+10	37	15
The Bank of Nova Scotia (BNS)	77	9.59E+10	4.05E+11	3.36E+10	9.73E+09	6.93E+09	42	13
Average							33x	14x
Median							35x	13x

TD latest balance sheet shows a basic EPS of 4.73, meaning that from all the business' earnings divided by the number of shares outstanding, each share gained that value. Considering the average multiple of P/E of the comparison companies, IB would project TD's share value by multiplying P/E by its current EPS. In this way, each TD share would be valued at 4.73*14 = 60.62 CAD. Additionally, considering the 33x average EV/EBITDA of the comparative companies allows us to utilize TD's 10,83 B EBITDA to arrive to a 357,39 B Enterprising value, that divided per share would price each at 204.22 CAD. This estimate would make the current price per share 63.23% undervalued.

The difference from valuation comes from the fact that EBITDA is always greater than net income since it represents exclusively the company's ability to generate cashflows. Therefore, IB has to interfere on the weight that each of these multiples have for the final valuation. This is achieved by taking in consideration the cashflow source of the target company, and how relevant it is to reduce interest, taxes, debt, and amortization for the firm's growth.

Conclusion

From these methods, the most often used in IB are relative valuations since it is easier to calculate and requires tangible data on the balance sheets, or income and cashflow statements, rather than derived percentages of growth. Additionally, it helps to not underprice a company, since it considers the current market of its competition. In certain cases, involving private companies, relative valuation might as well be the only possible way to determine the value of the company, given the lack of financial information available. However, relative valuations are particularly complicated for the matter of finding proper comparable companies since no firm is equal to the other. Therefore, finding valuation through comparison in this scenario could yield completely inaccurate multiples, and hence an inaccurate valuation. Moreover, even also a benefit, the overreliance on the present might cause the valuation method to overt irrational exuberance from the market.

Intrinsic valuation is technically the best method to evaluate a business. This since it does not account market phenomenon that might distort the true value of the company. Instead, it determines a present value that reflects the future capacity of the business to continue to make money, such as the DCF model displays. However, given the many estimates it assumes, the value obtain is very sensible to any inaccuracy in these. In fact, given the base of it is forecasting future performances, it is very hard to be completely sure about the projection of this method as it also does not account for changes in the capital structure or other probable events that might take place.

All of these considerations and more must be taken by IB firms when valuing companies for clients. Looking back at the example of TD, the DCF valuation outputted a 252 CAD intrinsic value. Whereas the same company was estimated to be worth 60.62 CAD using the P/E ratios, and 204.22 CAD using the EV/EBITDA multiple through relative valuation. As a matter of fact, it is a pre-establishment in IB that the highest valuation from a company will come from the PTA (due to the control premium), followed by DCF (due to bullish forecasts), and that the CCA will output the lowest value. It is therefore that IB firms either way calculate all of these since all of these values together draw circles closer and closer to the target firm's real value.

Sources

Adcock, B. (2013). Streetofwalls. Street Of Walls. https://www.streetofwalls.com/finance-training-courses/investment-banking-technical-training/valuation-techniques-overview/

EBITDA in private equity - measuring operating performance. CEPRES. (n.d.). https://cepres.com/insights/ebitda

Fernando, J. (n.d.). *What is present value? formula and calculation*. Investopedia. https://www.investopedia.com/terms/p/presentvalue.asp#:~:text=Present%20value%20(PV)%20is%20the,in%20strategic%20planning%20for%20businesses.

Hargrave, M. (2024, July 24). *Weighted average cost of capital (WACC) explained with formula and example*. Investopedia. https://www.investopedia.com/terms/w/wacc.asp

Pearl, J., & Rosenbaum, J. (2022). Investment banking, 3rd edition. Wiley.

Pignataro, P. (2013). Financial modeling and valuation: A practical guide to investment banking and private equity (1st ed.). Wiley.

Rupar Elia, A. (2024, April 16). *Levered free cash flow formula and calculation*. Agicap. https://agicap.com/en/article/levered-free-cash-flow/#:~:text=This%20measure%20is%20useful%20for,distribute%20cash%20to%20its%20investors.

Schmidt, J. (2023, December 22). *Valuation overview*. Corporate Finance Institute. https://corporatefinanceinstitute.com/resources/valuation/valuation/

Yahoo! (2024, December 19). *Bank of Montreal (BMO.TO) stock price, news, Quote & History*. Yahoo! Finance. https://finance.yahoo.com/quote/BMO.TO/

Yahoo! (2024, December 19). *Bank of Nova Scotia (BNS.TO) stock price, news, Quote & History*. Yahoo! Finance. https://ca.finance.yahoo.com/quote/BNS.TO/

Yahoo! (2024, December 19). *Canadian Imperial Bank of Commerce (CM.TO) stock price, news, Quote & History*. Yahoo! Finance. https://ca.finance.yahoo.com/quote/CM.TO/

Yahoo! (2024, December 19). *National Bank of Canada (NA.TO) stock price, news, Quote & History*. Yahoo! Finance. https://ca.finance.yahoo.com/quote/NA.TO/

Yahoo! (2024, December 19). *Royal Bank of Canada (RY.TO) stock price, news, Quote & History*. Yahoo! Finance. https://finance.yahoo.com/quote/RY.TO/

Yahoo! (2024, December 19). *The Toronto-Dominion Bank (TD.TO) stock price, news, Quote & History*. Yahoo! Finance. https://finance.yahoo.com/quote/TD.TO/